

Machine-Learning Detection of Potts-Model Phase Transitions: Finite-Size Scaling and Parameter Sensitivity*

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We investigate phase-transition behavior in the q -state Potts model using a range of machine-learning techniques, including dimensionality reduction, clustering, and neural-network-based representations. By constructing data-driven order parameters from learned representations, we identify the pseudo critical temperature and systematically examine finite-size scaling and parameter sensitivity.

I. MOTIVATION

A. Limitation of Ising Model

The Ising model is the cornerstone of statistical physics and phase transition theory, but it is also too “simple” sometimes and therefore lacks expressive power in many scenarios.

1. Limited state space

The Ising model restricts each spin to only two possible states, typically $s_i = \pm 1$. This binary description is too simplistic for many real physical systems that possess multiple internal states. For example, polycrystalline materials may involve grains with more than two orientational variants, adsorption on surfaces often occurs at multiple inequivalent sites, and multi-element alloys can have several competing local configurations. In such cases, the two-state Ising variable cannot faithfully represent the richness of local degrees of freedom.

2. Restriction to Z_2 symmetry

From a prospective of symmetry. Because the Ising model is based on a binary spin variable, its order parameter exhibits only Z_2 symmetry. This makes it unsuitable for systems with higher or continuous symmetries, such as q -state (multi-color) systems, clock models, or multistable systems with more than two equivalent ordered states. Phenomena where the symmetry of the order parameter plays a central role—such as symmetry breaking into multiple degenerate minima—cannot be fully captured within the Ising framework.

3. Inability to generically describe first-order phase transitions

In its simplest form with short-range interactions and no additional fields, the Ising model typically exhibits continuous (second-order) phase transitions rather than first-order ones. As a result, it fails to naturally describe systems where phase transitions involve latent heat, phase coexistence, and discontinuous jumps of the order parameter.

B. Pros and Cons of Different Algorithms in Studying Phase Transition Problems.

Nowadays, there are plenty of ML algorithm to choose to explore phase transition behavior. Including supervised learning, unsupervised learning and generative model. We will investigate the pros and cons of different algorithms in studying phase transition problems.

II. BRIEF INTRODUCTION TO POTTS MODEL

To deal with the limitation of Ising model, we need to introduce Potts model[1], which is a natural generalization to Ising model.

The Potts model is a natural generalization of the Ising model, in which each lattice site can take one of q discrete states instead of only two.

$$s_i \in \{1, 2, \dots, q\}$$

These states are often interpreted as different “colors,” and neighboring sites interact by favoring the same state. When $q = 2$, the Potts model reduces exactly to the Ising model, making it a unified framework for studying systems with discrete symmetries of varying complexity.

The Hamiltonian of Potts model is defined as:

$$H = -J \sum_{\langle i,j \rangle} \delta_{s_i s_j} \quad (1)$$

Here, $\langle \rangle$ means sum over neighbors, δ_{s_i, s_j} is the Kronecker delta and $J > 0$ denotes the ferromagnetic coupling.

* A footnote to the article title

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Because it allows multiple local states, the Potts model is well suited for describing multistate physical systems, such as polycrystalline grain orientations, adsorption on surfaces with several equivalent binding sites, and multi-component alloys. Its order parameter exhibits a Z_q symmetry, enabling the study of symmetry breaking into more than two degenerate ordered phases—something the Ising model cannot capture.

An important feature of the Potts model is the rich variety of phase-transition behaviors it exhibits. Depending on the dimension and the value of q , the model can show either continuous (second-order) or discontinuous (first-order) phase transitions. For example, in two dimensions the transition is second order for small q but becomes first order for sufficiently large q , making the Potts model a classic theoretical laboratory for studying the nature of phase transitions and universality.

A typical Potts model is shown like Figure 1.

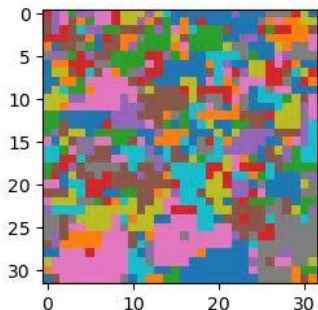


FIG. 1. A typical Potts model (different colors stands for different states)

III. SAMPLE GENERATOR: SW ALGORITHM

A. Obstacles of Metropolis Algorithm

To generate equilibrium configurations of the q -state Potts model, a straightforward approach is the Metropolis Monte Carlo algorithm based on local spin updates. The Hamiltonian of the Potts model is given by

$$H = -J \sum_{\langle i,j \rangle} \delta_{s_i, s_j}, \quad s_i \in \{1, 2, \dots, q\},$$

In the Metropolis scheme, a single lattice site is selected and its state is proposed to change to another color, which is accepted with probability

$$P_{\text{acc}} = \min(1, e^{-\beta \Delta E}). \quad (2)$$

However, for the Potts model this local-update strategy becomes inefficient at low temperatures and near the critical point. Large domains of identical colors are formed, and flipping a single spin typically breaks many satisfied bonds, leading to a large energy increase and a very low acceptance rate. As a consequence, the simulation suffers from severe critical slowing down, characterized by

long autocorrelation times and poor sampling of different symmetry-broken states.

B. Swendsen–Wang Cluster Algorithm

To overcome these limitations, we employ the Swendsen–Wang (SW) cluster algorithm, which performs non-local updates by flipping entire clusters of spins simultaneously [2]. The algorithm proceeds as follows. For each nearest-neighbor pair (i, j) with identical states $s_i = s_j$, a bond is activated between them with probability

$$p = 1 - e^{-\beta J}. \quad (3)$$

No bonds are placed between spins of different states. The activated bonds partition the lattice into disconnected clusters.

Once the clusters are identified, each cluster is independently assigned a new spin state, randomly chosen from the q possible colors with equal probability. All spins within the same cluster are updated simultaneously. This procedure satisfies detailed balance and ergodicity, and can be formally understood through the Fortuin–Kasteleyn random-cluster representation of the Potts model.

C. Advantages of the S–W Algorithm

The key advantage of the SW algorithm lies in its ability to update correlated regions collectively. Since entire clusters are flipped at once, the updates do not incur the large energy penalties typical of single-spin flips in ordered phases. This dramatically reduces critical slowing down and leads to much shorter autocorrelation times, especially near the phase transition.

Moreover, the SW algorithm allows efficient transitions between different symmetry-related ordered states, which is crucial for accurately sampling the multi-degenerate free-energy landscape of the Potts model. For these reasons, the Swendsen–Wang algorithm is particularly well suited as a configuration generator in studies of phase transitions and in data-driven analyses based on Potts model samples.

IV. DETERMINE PHASE TRANSITION ORDER

A major difference between the Potts model and the Ising model is that, by increasing the number of states (from 2 to q), the Potts model exhibits a richer variety of phase transition orders. In the standard Ising model, the phase transition is always second-order. In contrast, theory tells us that the order of the phase transition in the Potts model depends on the value of q . Specifically,

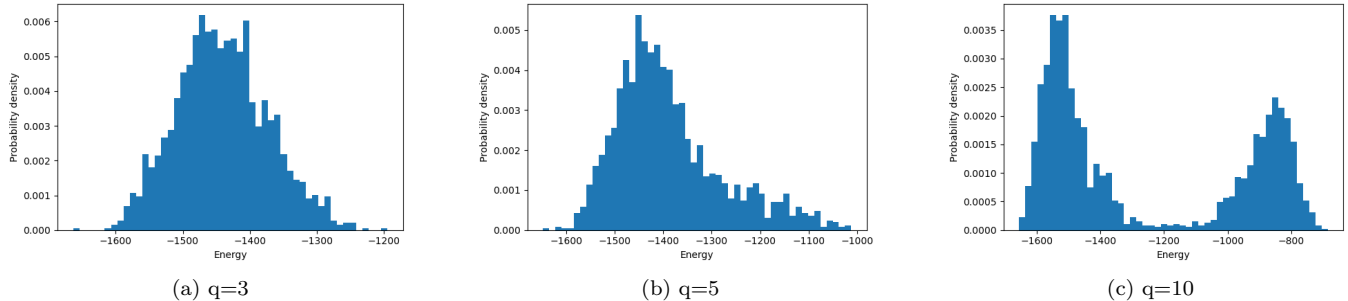


FIG. 2. The energy distribution at T_C of different q . We take 2000 samples for each situation and divide them into 50 bins. We choose $L = 30$.

for the two-dimensional Potts model, the transition is second-order when $q \leq 4$, and becomes first-order when $q > 4$.

In two dimensions, the Potts model exhibits a phase transition at a critical temperature[1]

$$T_C = \frac{1}{\ln(1 + \sqrt{q})}, \quad (4)$$

with the nature of the transition depending sensitively on the number of states q .

a. Free energy and definition of transition order. The order of a phase transition is determined by the analyticity of the free energy $F(T)$. A first-order phase transition is characterized by a discontinuity in the first derivative of the free energy,

$$E = -\frac{\partial F}{\partial T}, \quad (5)$$

implying a finite latent heat. In contrast, a second-order (continuous) phase transition exhibits continuous first derivatives, while second derivatives such as the specific heat

$$C = -T \frac{\partial^2 F}{\partial T^2} \quad (6)$$

diverge or become non-analytic at the critical point.

b. Entropy-energy competition and large- q behavior. The key mechanism driving the change in transition order in the Potts model is the competition between energy and entropy in the free energy

$$F = E - TS. \quad (7)$$

At low temperatures, the system minimizes energy by selecting one of the q symmetry-broken ordered states, resulting in a low-entropy phase. At high temperatures, all q spin states are approximately equally populated, yielding a disordered phase with entropy scaling as

$$S_{\text{dis}} \sim N \ln q. \quad (8)$$

As q increases, the entropy gain of the disordered phase grows rapidly, enhancing the free-energy difference between the ordered and disordered phases. Consequently, intermediate partially ordered configurations

become thermodynamically unfavorable, leading to a discontinuous jump between the two phases.

c. Landau theory perspective. This behavior can be further understood within a Landau-type effective free energy expansion. For the Potts model, the order parameter is a $(q-1)$ -dimensional vector $\vec{m}$, and the symmetry allows for a cubic invariant when $q \geq 3$. The effective free energy may be written as

$$\mathcal{F}(\vec{m}) = a(T)m^2 + bm^3 + cm^4 + \dots, \quad (9)$$

where $m = |\vec{m}|$. The presence of the cubic term generically induces a first-order phase transition. In two dimensions, thermal fluctuations suppress this effect for $q \leq 4$, resulting in a continuous transition, whereas for $q \geq 5$ the cubic term dominates, yielding a genuine first-order transition. This explains the well-known change in universality at $q = 4$.

d. Configuration-space picture. From the viewpoint of configuration space, a second-order transition corresponds to a single free-energy minimum that deforms continuously across T_C , while a first-order transition exhibits multiple competing minima with comparable free energies at T_C . The latter scenario leads to phase coexistence and bimodal energy distributions, which are symbols of first-order behavior. We plot the energy distribution at critical temperature in Figure 2. The unimodal distribution for $q = 3$ and the bimodal distribution for $q = 10$ are in excellent agreement with our theoretical expectations. We also show the energy distribution at the transitional value $q = 5$, which clearly exhibits an intermediate state between the unimodal and bimodal regimes.

V. DETERMINATION OF CRITICAL TEMPERATURE

A. PCA + K-means

We first apply principal component analysis (PCA) as a linear, unsupervised dimensionality-reduction technique to investigate whether signatures of the phase transition in the Potts model can be extracted directly from

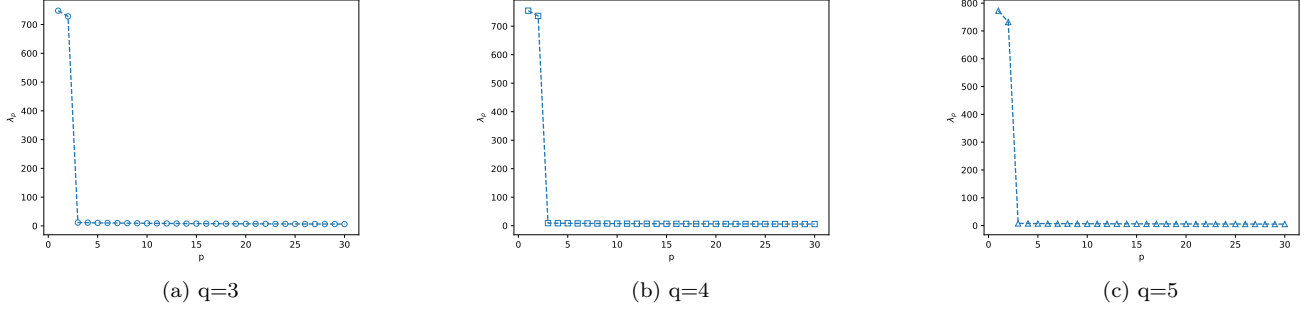


FIG. 3. The PCA eigenvalues for different q -state Potts model with all samples ($L=40$)

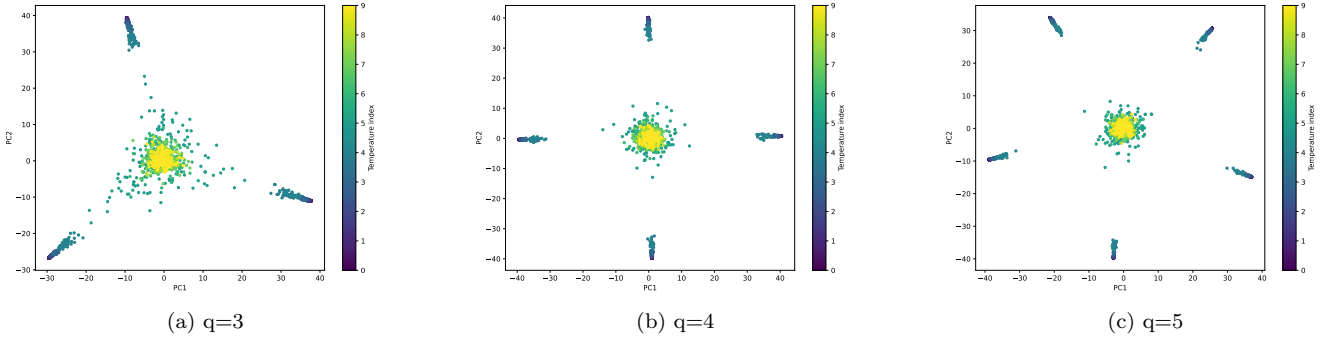


FIG. 4. PCA projection of embedded configurations of the 2D q -state Potts model onto the first two principal components ($L=40$). Each point corresponds to one configuration, colored by the temperature index t (ordered from low to high temperature)

configuration data [3]. Each configuration is mapped to a high-dimensional feature vector using a one-hot embedding of the spin states, and PCA is performed on the pooled dataset collected across temperatures.

PCA identifies orthogonal directions in configuration space that capture the largest variance in the data. We plotted the eigenvalues λ_p corresponding to the first thirty principal components, as shown in Fig. 3. Obviously, the eigenvalues of the first two principal components are much larger than those of the subsequent ones, containing the vast majority of the information. Therefore, we will conduct our analysis within the parameter space of the first two principal components, ignoring the information from the higher-order principal components.

Although PCA is agnostic to temperature labels and physical symmetries, the leading principal components still encode meaningful information about the phase structure. In the low-temperature regime, projected configurations form q well-separated clusters in the plane spanned by the first two principal components, reflecting the q -fold degeneracy of the ordered phase, as shown in Fig. 4. As temperature increases, these clusters approach each other and eventually merge into a single cloud at high temperature, corresponding to the disordered phase.

To quantify this geometric evolution, we use the PCA projection and apply k -means clustering with $k = q$ clusters at each temperature [4]. From the resulting cluster

centers, we define a data-driven order parameter $m(T)$ based on the separation of cluster centers in PCA space. Specifically, let $\{\mathbf{c}_k(T)\}_{k=1}^q$ denote the q cluster centers at temperature T , then we define

$$m(T) = \frac{1}{q} \sum_{k=1}^q \|\mathbf{c}_k(T)\|_2. \quad (10)$$

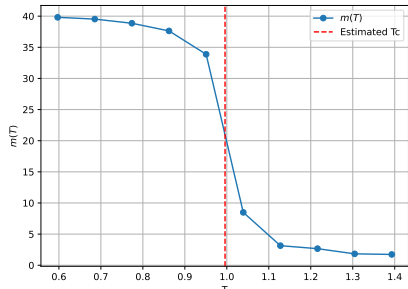
This is not the magnetization m of the Potts model, but rather a proxy of geometrical order on the PCA plane. It is large when the q clusters are well separated (ordered phase) and decreases rapidly as the system approaches the transition, saturating at a small value in the disordered phase.

Next, we fit the temperature dependence of the order parameter using a spline interpolation. We define the pseudo-critical temperature $T_c^{(L)}$ as the point where the first derivative of the fitted curve reaches its minimum, i.e., where the order parameter exhibits the steepest drop:

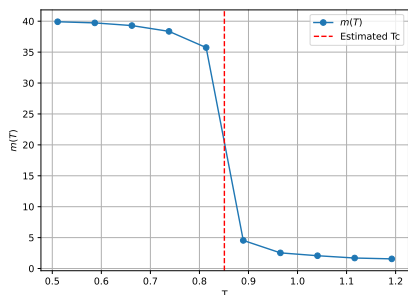
$$T_c^{(L)} = \arg \min_T \frac{dm(T)}{dT}. \quad (11)$$

For systems with $q=3$ and $q=5$, the results are shown in Fig. 5. Compared with the theoretical values $T_c(q=3) \approx 0.99497$ and $T_c(q=5) \approx 0.85153$ calculated by

Eq.(4), the critical temperature obtained by this method is quite accurate. The errors here come from two sources: the random error in generating the samples, and the limitation of the method by the finite-size scaling, which we will discuss later.



(a) $q=3$, $L=40$, $T_c^{(L)} \approx 0.99577$



(b) $q=5$, $L=40$, $T_c^{(L)} \approx 0.85085$

FIG. 5. The order parameter changes with temperature and the corresponding pseudo-critical temperature (spline fitting results are not shown).

B. CNN

1. Theory

In classical Potts model, phase transitions are traditionally characterized by the emergence or disappearance of an order parameter associated with symmetry breaking. For the q -state Potts model, the low-temperature phase exhibits spontaneous breaking of the discrete permutation symmetry among the q spin states, while the high-temperature phase is disordered.

From a machine learning perspective, supervised learning has been widely applied to classify configurations sampled at different temperatures into ordered and disordered phases. In this framework, spin configurations are used as inputs, and phase labels—typically assigned according to whether the temperature is below or above the critical temperature T_C —are used as supervision signals. Neural networks trained in this manner often display a sharp change in classification confidence near the phase

transition, which can be used to estimate T_C .

An important conceptual question is what physical quantity such a supervised model actually learns. In systems where a well-defined order parameter is already known, it is natural to ask whether the neural network effectively reconstructs the conventional order parameter through a nonlinear mapping.

In this work, we view supervised learning as an automated procedure for extracting an *effective order parameter* from raw microscopic configurations. Given a spin configuration $\{\sigma_i\}$, the trained convolutional neural network[5] (CNN) defines a scalar function

$$P_{\text{CNN}}(\{\sigma_i\}) \in [0, 1], \quad (12)$$

which represents the predicted probability that the configuration belongs to the ordered phase. If this output is strongly correlated with the physical order parameter, then the neural network does not introduce a new phase descriptor but instead learns a nonlinear reparameterization of an existing one.

For the q -state Potts model, a natural choice of order parameter is the generalized magnetization[6]

$$m = \frac{q \max_k n_k - N}{N(q-1)}, \quad (13)$$

where n_k denotes the number of spins in state k , and N is the total number of spins. This definition treats the q ordered states symmetrically and interpolates smoothly between $m \approx 0$ in the disordered phase and $m \approx 1$ in the fully ordered phase.

Under this interpretation, the central theoretical question addressed by the CNN analysis can be formulated as follows: Is the CNN output P_{CNN} effectively a monotonic function of the magnetization m ? Demonstrating such a relationship would provide direct evidence that supervised learning, when applied to a system with a known order parameter, primarily serves as a data-driven mechanism for reconstructing that order parameter rather than uncovering fundamentally new physics.

2. Algorithmic implementation

To test the theoretical interpretation outlined above, we implement a supervised convolutional neural network to analyze configurations of the Potts model. Configurations are sampled across a range of temperatures spanning both sides of the theoretical critical temperature T_C .

Each spin configuration of size $L \times L$ is encoded using a one-hot representation with q channels, resulting in an input tensor of shape (q, L, L) . This representation preserves the discrete nature of the Potts variables while allowing convolutional filters to capture spatial correlations and domain structures.

The CNN architecture consists of two convolutional layers with rectified linear unit (ReLU) activations, followed by global average pooling and fully connected layers. The convolution kernel size are chosen to be 3×3 ,

which exactly corresponds to the local structure involving nearest neighbors plus next-nearest neighbors. The final layer employs a sigmoid activation function, producing a scalar output interpreted as the probability that the input configuration belongs to the ordered phase.

To ensure that the network learns robust phase-specific features, the training data are intentionally chosen from temperatures well below and well above the critical temperature. Configurations near the critical region are excluded from training and reserved exclusively for testing.

After training, the CNN is evaluated on configurations sampled at many intermediate temperatures. For each temperature, the average CNN output $\langle P_{\text{CNN}}(T) \rangle$ is computed. The temperature at which this quantity changes most rapidly is identified as the CNN-inferred critical temperature.

To directly probe the physical content of the learned representation, the magnetization m is computed for each test configuration and compared with the corresponding CNN output. Furthermore, by averaging the CNN output within bins of magnetization, we plot their relationship.

3. Results

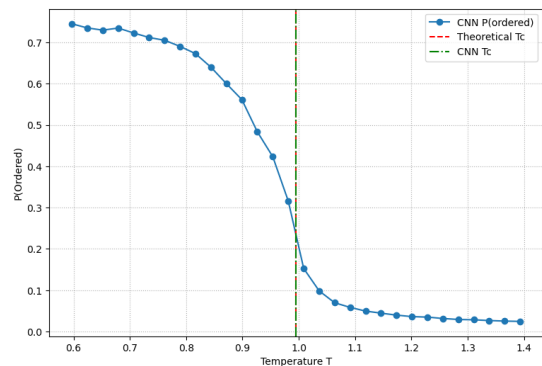
a. Critical Temperature First, we determine the critical temperature T_C by plotting the curve of order parameter vs temperature. The results are shown in Figure 6. We can see that at small q , the result from CNN matches the theory pretty well. However, with the growing of q , CNN critical temperature deviate from theoretical value.

The underlying physical reason is that the Potts model with large q exhibits a strongly first-order phase transition. Although, in theory, the two-dimensional Potts model undergoes a first-order transition for all $q \geq 5$, finite-size scaling effects prevent the order parameter vs. temperature curve from becoming a perfect step function; instead, it takes on a sigmoid-like shape. However, as q increases, the discontinuity associated with the first-order transition becomes increasingly pronounced.

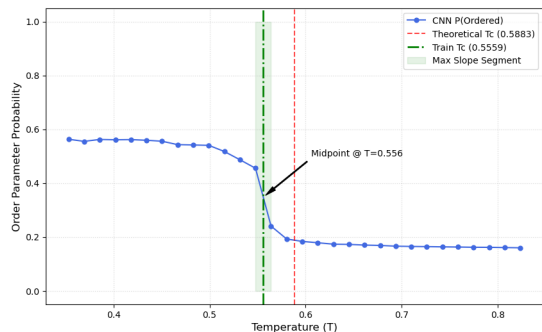
Our CNN identifies the critical point as the temperature where the slope of the order parameter curve is maximal. Yet, for large q , there is no broad critical region with a smooth slope for the CNN to learn from. At $T = T_C$, the system simultaneously hosts fully ordered states, fully disordered states, and mixed states separated by large domain walls. This coexistence of multiple distinct phases makes classification inherently difficult.

In contrast, for small q , the CNN effectively learns the continuous evolution of local correlation lengths and domain sizes—a “differentiable” geometric feature problem. Consequently, the curve of $P(\text{ordered})$ vs. T is a well-behaved S-shaped sigmoid, and the prediction of T_C is highly accurate.

When q is sufficiently large, however, the CNN no longer learns the presence of spontaneous symmetry



(a) $q=3$, $L=30$



(b) $q=20$, $L=30$

FIG. 6. Order parameter vs temperature. Red line stands for theoretical prediction while the green line for CNN determination.

breaking; instead, it learns whether a single color dominates overwhelmingly. Furthermore, because we use binary labels (ordered/disordered), and such labels become ill-defined near T_C in a first-order transition (where both phases coexist), the CNN is forced to shift its predicted T_C toward temperatures where the training data are more clearly distinguishable—typically toward the temperature where metastable states disappear—rather than the true thermodynamic critical temperature T_C .

b. Order Parameter Second, we investigate the relation between order parameter of CNN (i.e. probability P_{CNN}) and physical order parameter (i.e. magnetization m). We plot the graph of $P_{\text{CNN}} - m$ in Figure 7. These scatter plots of P_{CNN} versus m reveals a strong, monotonic correlation between the two quantities. Configurations with small magnetization consistently yield CNN outputs close to zero, while highly magnetized configurations yield outputs close to one.

From these two figures, we can also discern the difference in the order of the phase transition. Observe the data points near the critical temperature T_c ($T_c = 0.995$ for $q = 3$; $T_c = 0.701$ for $q = 10$). We find that, for $q = 3$, the points are tightly clustered in a single region, whereas for $q = 10$, they split and reside at opposite ends of the data band. This bimodal separation is a hallmark of the discontinuous jump characteristic of a first-order

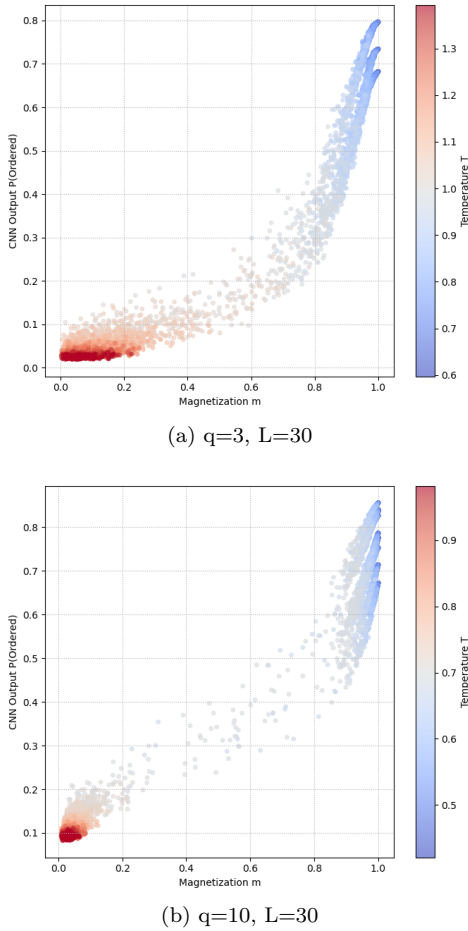


FIG. 7. CNN output P vs magnetization m

transition, in contrast to the continuous behavior of a second-order transition. Consequently, for $q = 3$, the data band appears as a continuous distribution, while for $q = 10$, a sparse region emerges in between, signaling the abrupt jump in the order parameter.

To better illustrate this change, we averaged the data points and plotted their overall trend as a line graph (Figure 8). For $q = 3$, a smooth dependence of $\langle P_{\text{CNN}} \rangle$ on m is observed. While for $q = 10$, it is smooth at both ends but violently fluctuates near critical point. This demonstrates that the CNN output can be interpreted as a nonlinear function of the physical order parameter.

These results provide direct numerical evidence that, for the Potts model where a conventional order parameter is known, supervised learning does not uncover an independent phase descriptor. Instead, the CNN effectively reconstructs the magnetization from raw spin configurations and uses it as the basis for phase classification. This clarifies the role of supervised learning in statistical physics as a flexible, data-driven tool for extracting effective order parameters, particularly valuable in systems where such quantities are unknown or difficult to define analytically.

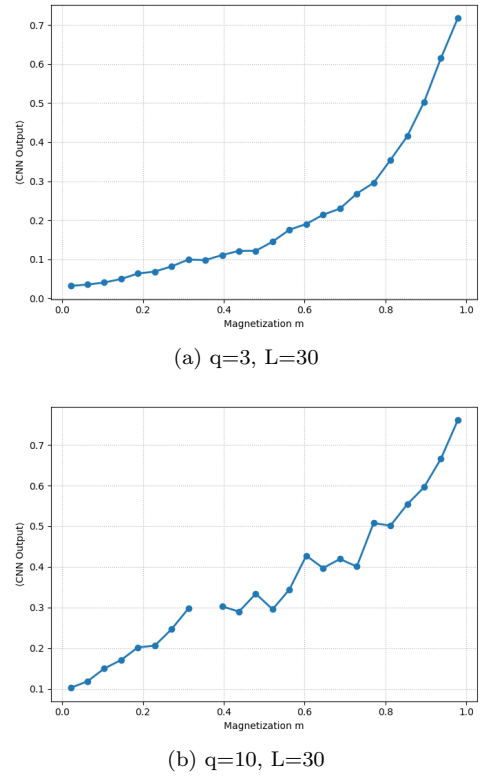


FIG. 8. Averaged CNN output P vs magnetization m

VI. FINITE-SIZE SCALING

In numerical studies of phase transitions, simulations are necessarily performed on finite lattices, leading to systematic deviations from thermodynamic-limit behavior. These finite-size scaling arise because singularities in thermodynamic observables are rounded and shifted when the system size L is finite. As a result, quantities that exhibit nonanalytic behavior at the critical temperature T_c in the infinite-volume limit instead show smooth crossovers at a pseudo-critical temperature $T_c(L)$, which generally depends on the lattice size.

In the context of machine-learning-based approaches, finite-size scaling manifest in a geometric manner. Configuration distributions at finite L do not collapse sharply from ordered to disordered phases but instead evolve continuously with temperature. This smoothing affects both linear and nonlinear representations of the data and can bias the location of the transition inferred from clustering, dimensional reduction, or latent-space observables.

Within the PCA-based analysis, finite-size scaling are clearly visible. Although the leading principal components separate ordered and disordered configurations, the merging of clusters in PCA space occurs over a finite temperature window rather than at a single point. Consequently, the data-driven order parameter $m(T)$ exhibits a rounded crossover, and the estimated pseudo-critical temperature depends weakly but systematically on L . This behavior is consistent with the known finite-size

rounding of conventional order parameters in the Potts model.

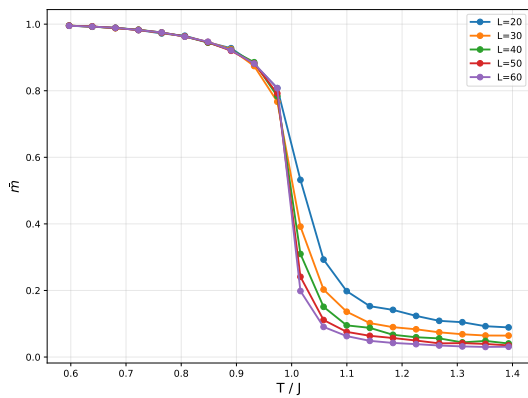


FIG. 9. Temperature dependence of the order parameter $\bar{m}$ as a function of the temperature T/J ($L = 20, 30, 40, 50, 60$). As the system size increases, the change in the order parameter near the critical temperature becomes steeper, and it more closely approximates the abrupt change in the thermodynamic limit.

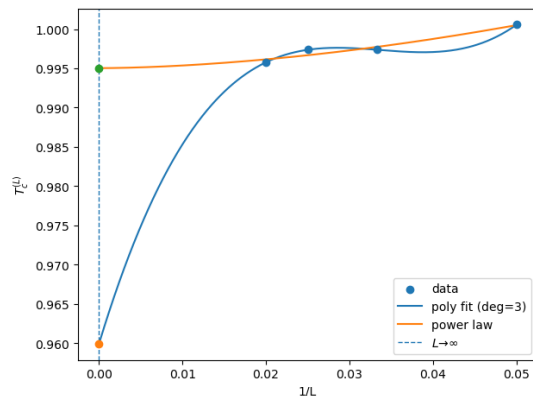


FIG. 10. Finite-size extrapolation of the pseudo-critical temperature $T_c^{(L)}$ as a function of $1/L$. Markers denote $T_c^{(L)}$ extracted for each system size L from the steepest-drop criterion of the order-parameter curve (i.e., the minimum of the first derivative after spline fitting). The blue curve shows a third-order polynomial fit, while the orange curve shows a power-law fit used to capture the expected finite-size scaling behavior. The vertical dashed line indicates the thermodynamic-limit axis $1/L \rightarrow 0$ ($L \rightarrow \infty$); the intercepts of the fitted curves at $1/L = 0$ provide an estimate of the critical temperature T_c in the thermodynamic limit. Here $T_c(\text{poly}) \approx 0.95988$ and $T_c(\text{power}) \approx 0.99503$.

The K-means clustering procedure inherits this size dependence, because it relies on the geometry of the PCA embedding, as shown in the Fig. 9. While the qualitative behavior of $\bar{m}(T) = m(T)/L$ is robust across different lattice sizes, the location and sharpness of the crossover vary with L , especially for continuous transitions and

weakly first-order transitions. These observations indicate that linear dimensional-reduction methods alone are insufficient to fully suppress finite-size scaling.

Based on PCA and k -means, to address the limitations of the effective size effect, we can extrapolate to the case of $L \rightarrow \infty$ through fitting. Here, we employ polynomial fitting and power fitting, with the results for $q=3$ shown in Fig. 10. It can be seen that the power fitting result is closer to the theoretical value than before, but this method still has some error.

VII. TOPOLOGICAL DATA ANALYSIS

Topological data analysis (TDA) provides a set of tools to characterize the “shape” of data in a manner that is robust to noise and insensitive to continuous deformations [7][8]. In contrast to approaches that rely on explicit coordinates in a low-dimensional embedding (e.g., PCA) or model-dependent latent variables (e.g., VAE), TDA summarizes datasets through topological invariants that capture global structural properties such as connectivity and loop-like patterns. This makes TDA particularly attractive for studying phase transitions, where the underlying configuration distribution can reorganize qualitatively as temperature crosses the critical region.

A standard TDA pipeline begins with a point cloud $X = \{x_n\}_{n=1}^N$ in some metric space. One then builds a family of simplicial complexes parameterized by a scale parameter ϵ , typically through a Vietoris–Rips construction. As ϵ increases, isolated points begin to connect, connected components merge, and higher-dimensional simplices fill in. Persistent homology tracks the appearance (birth) and disappearance (death) of topological features across scales. The output is a persistence diagram, which is a multiset of points (b, d) , where b and d denote the birth and death scales of a feature. In this work, we focus on H_0 , corresponding to connected components.

Persistence diagrams provide a compact summary of multi-scale topology and can be compared across datasets via diagram distances such as the Wasserstein distance.

We apply TDA to ensembles of Monte Carlo configurations of the two-dimensional q -state Potts model. For each temperature T_i in a predefined grid $\{T_i\}_{i=1}^{n_T}$, we generate n_{samples} independent configurations using the Swendsen–Wang cluster algorithm. Each configuration is mapped to a high-dimensional feature vector using a one-hot embedding of spin states. Concretely, if $s(\mathbf{r}) \in \{0, 1, \dots, q-1\}$ denotes the Potts spin at lattice site $\mathbf{r}$, we represent it by a one-hot vector in $\mathbb{R}^q$, and then flatten the $L \times L$ lattice into a single vector. This yields a point cloud $X_i = \{x_{i,n}\}_{n=1}^{n_{\text{samples}}}$ at each temperature T_i .

For each X_i , we compute persistence diagrams using a Vietoris–Rips filtration. In practice, we compute H_0 persistence diagrams. We then quantify the dissimilarity between temperatures by computing the Wasserstein distance between their persistence diagrams. Let $D_0(X_i)$

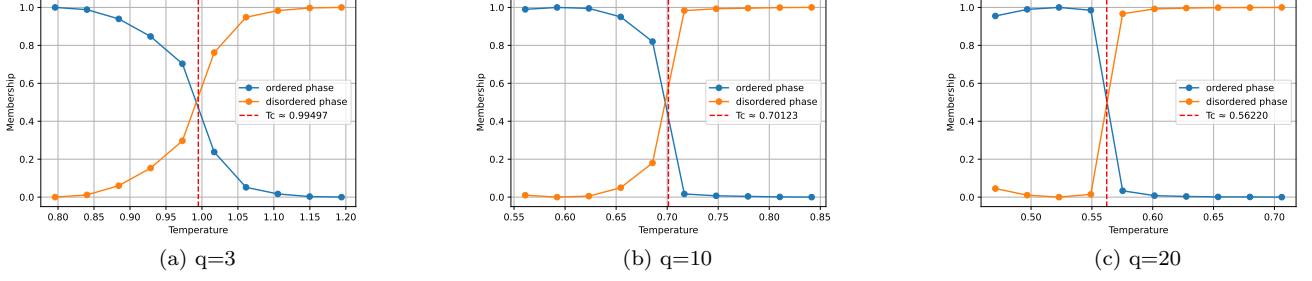


FIG. 11. The critical temperature ($N=150$) obtained using TDA when $q = 3, 10, 20$ shows that the system exhibits a first-order phase transition when q is sufficiently large. However, a low sample size can cause the critical temperature to deviate from the expected value.

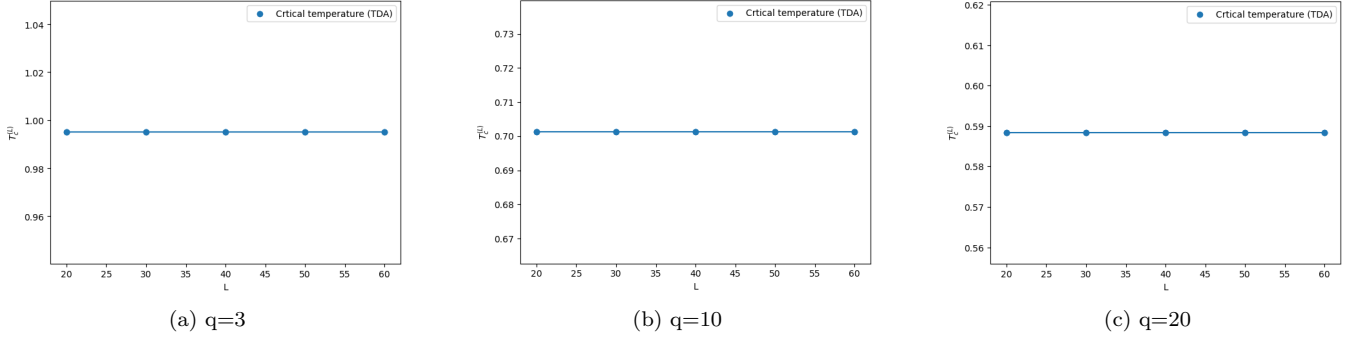


FIG. 12. With a suitable sample size, the critical temperature calculated by TDA is almost independent of L , which shows that TDA can indeed effectively avoid the limitations of finite-size scaling.

denote the H_0 diagrams of X_i . We define an inter-temperature distance

$$D_{ij} = W(\mathcal{D}_0(X_i), \mathcal{D}_0(X_j)), \quad (14)$$

where $W(\cdot, \cdot)$ denotes the Wasserstein distance.

The matrix $D \in \mathbb{R}^{n_T \times n_T}$ is converted into a similarity kernel using a Gaussian radial basis function (RBF),

$$K_{ij} = \exp\left(-\frac{D_{ij}^2}{2\sigma^2}\right), \quad (15)$$

where σ is a kernel bandwidth parameter. To reduce sensitivity to outliers and to improve robustness across lattice sizes, we set σ using a robust statistic of pairwise distances (e.g., the median of $\{D_{ij}\}_{i < j}$), rather than the mean. We then perform spectral decomposition on K to obtain its eigenvalues and eigenvectors. The leading eigenvector corresponds to a nearly constant mode, while the second eigenvector encodes the dominant nontrivial partitioning structure on the temperature graph.

Following the spectral-clustering interpretation, we use the second eigenvector v_2 to define a continuous membership function that separates the ordered and disordered regimes. After fixing the sign ambiguity of v_2 such that

low-temperature points have larger membership, we normalize it to $[0, 1]$:

$$m_A(T_i) = \frac{v_2(T_i) - \min_j v_2(T_j)}{\max_j v_2(T_j) - \min_j v_2(T_j)}, \quad (16)$$

$$m_B(T_i) = 1 - m_A(T_i), \quad (17)$$

where m_A and m_B can be interpreted as memberships of the ordered-like and disordered-like phases, respectively. The pseudo-critical temperature can be estimated from the crossing of the memberships, i.e., the temperature where $m_A(T) \approx 0.5$. In practice, we locate the crossing interval on the discrete temperature grid and take its midpoint as an estimate, with an uncertainty set by the temperature step size.

Applying the above TDA pipeline, we obtain a sharp membership transition across temperature. For moderate q (e.g., $q = 3$ and $q = 10$), the inferred pseudo-critical temperature agrees closely with the exact critical temperature, as shown in Fig. 11. And, importantly, exhibits negligible dependence on the lattice size L over the range we tested. This behavior contrasts with PCA-based geometric order parameters, which show visible finite-size rounding and a systematic drift of $T_c(L)$. The reduced L sensitivity in TDA suggests that persistence-diagram distances capture global structural changes in configuration

space that remain stable under finite-size fluctuations.

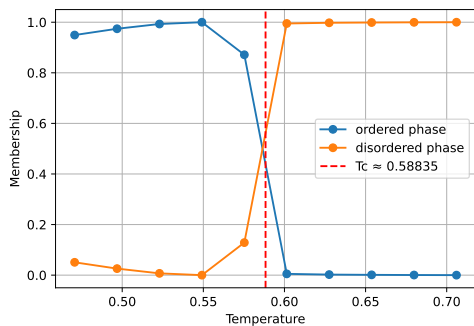


FIG. 13. When $q = 20$, after increasing the number of samples for each temperature to 800, the calculated critical temperature is consistent with the theoretical value.

For large q (e.g., $q = 20$), where the two-dimensional Potts model undergoes a strongly first-order phase transition, we observe that the accuracy of the TDA estimate becomes sensitive to the effective diversity of samples collected near T_c . With insufficient sampling, trajectories can remain trapped in metastable phases, leading to persistence diagrams that represent only one branch of the coexisting distributions and thus biasing the inter-temperature distance matrix. Increasing the number of sampled configurations, or combining ensembles generated from distinct initial conditions (ordered and disordered), substantially improves agreement with the theoretical T_c . These results indicate that, while the TDA representation itself is robust to finite-size scaling, accurate critical-temperature estimation in strongly first-order regimes still requires adequate sampling of phase coexistence.

To sum up, TDA provides an effective, representation-agnostic framework for detecting phase transitions directly from configuration data. By comparing temperatures through persistence-diagram distances and extracting the dominant partition structure via spectral analysis, our method yields accurate estimates of T_c with significantly reduced finite-size dependence for moderate q , while also clarifying the sampling requirements for strongly first-order transitions at large q .

VIII. CONCLUSION

In this work, we investigated phase-transition behavior in the two-dimensional q -state Potts model through a suite of machine-learning and geometry-driven techniques, with an emphasis on finite-size effects and parameter sensitivity. Equilibrium configurations were generated using the Swendsen–Wang cluster algorithm, which effectively mitigates critical slowing down and enables reliable sampling across a broad temperature range.

We first introduced a fully unsupervised,

representation-based approach using PCA on one-hot embedded configurations. The leading principal components capture the dominant variance in configuration space and reveal a clear geometric separation between ordered and disordered regimes: at low temperature, samples form q symmetry-related clusters in the (PC1, PC2) plane, which merge into a single cloud as temperature increases. Building on this observation, we defined a data-driven order parameter $m(T)$ from the average distance of q K-means cluster centers to the origin. The pseudo-critical temperature $T_c^{(L)}$ was then determined by the steepest-drop criterion, i.e., the minimum of the first derivative of a spline fit to $\bar{m}(T)$. This pipeline yields accurate critical-temperature estimates for moderate q , but also exhibits the expected finite-size rounding and a systematic L -dependence of $T_c^{(L)}$, motivating an explicit finite-size extrapolation. Polynomial and power-law fits of $T_c^{(L)}$ versus $1/L$ partially reduce this bias, with power-law extrapolation providing improved agreement with the exact T_c for the representative case $q = 3$.

We further examined a supervised CNN classifier as an automated mechanism for extracting an effective order parameter from raw configurations. By comparing the CNN output P_{CNN} with the physical magnetization m , we found a strong monotonic correlation, supporting the interpretation that the network primarily reconstructs (a nonlinear transform of) the conventional order parameter rather than discovering an independent phase descriptor. Importantly, we observed that the CNN-inferred transition temperature deviates increasingly from the theoretical value as q grows and the transition becomes strongly first order, consistent with phase coexistence and metastability near T_c that complicate binary supervision and blur the notion of a single “most-informative” critical region for classification.

Finally, we implemented a topological data analysis (TDA) pipeline based on persistent homology. By computing H_0 persistence diagrams for temperature-resolved point clouds, measuring inter-temperature dissimilarity via Wasserstein distances, and extracting the dominant partition structure through spectral analysis of an RBF kernel (with a robust bandwidth choice), we obtained a continuous membership function whose crossing provides an estimate of T_c . For moderate q (e.g., $q = 3$ and $q = 10$), this TDA-based estimate is both accurate and nearly independent of system size over the tested range, highlighting that persistence-diagram distances capture global structural changes that remain stable under finite-size fluctuations. For large q in the strongly first-order regime (e.g., $q = 20$), we found that accuracy becomes sensitive to sample diversity near T_c ; increasing the number of configurations per temperature and/or combining ensembles initialized from distinct phases substantially improves agreement with theory.

Overall, our results delineate complementary strengths and limitations of PCA-based geometry, supervised

CNNs, and TDA for phase-transition detection in the Potts model. Linear dimensional reduction offers an interpretable geometric picture but retains visible finite-size drift; supervised learning can effectively reconstruct known order parameters yet may be biased in strongly first-order settings; and TDA provides a representation-agnostic route that significantly suppresses finite-size de-

pendence for moderate q , while still requiring adequate sampling to resolve phase coexistence at large q . These findings provide a practical guide for selecting and validating ML-based diagnostics of critical phenomena, especially when the nature of the transition and finite-size effects must be carefully controlled.

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